

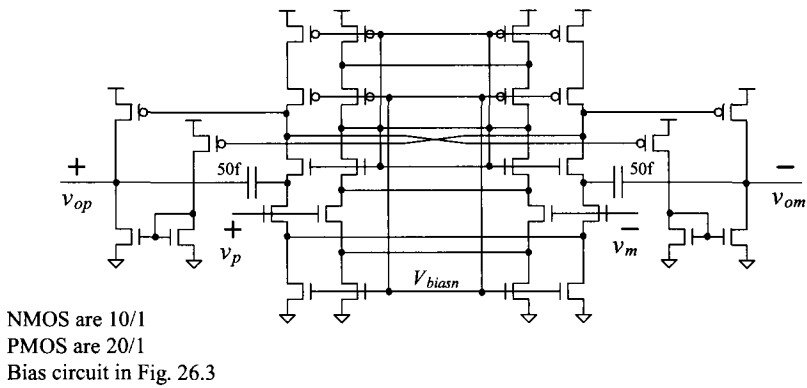


Figure 26.22 Basic two-stage op-amp without CMFB.

Figure 26.23 shows the DC characteristics of the op-amp in Fig. 26.22 where, once again, we've held the inverting op-amp input at the common-mode voltage and swept the voltage on the noninverting input. Notice how the outputs swing all the way from ground to V_{DD} ($= 1\text{V}$). Further notice how the differential output voltage swings from -1 to $+1\text{V}$ (a doubling in the output swing as discussed in Sec. 25.2.) The op-amp's DC gain is approximately 500 without a DC load. Notice how the two op-amp outputs cross at 400 mV (not at V_{CM} where they should). We'll discuss the CMFB in a moment.

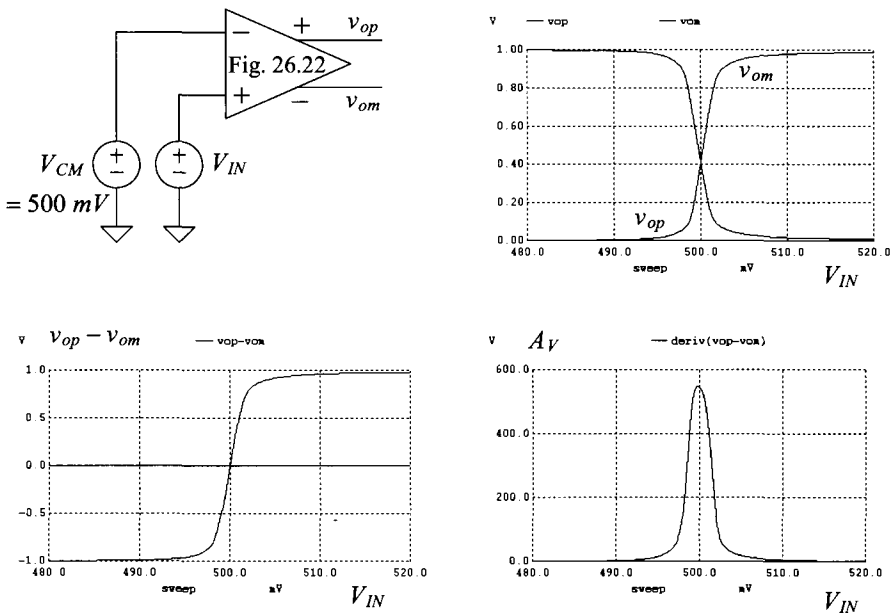


Figure 26.23 DC behavior and gain of the op-amp in Fig. 26.22.

Step Response

To determine both the stability of the op-amp in Fig. 26.22 and the settling time under certain loading conditions, consider the configuration seen in Fig. 26.24. We used relatively small resistors, 20k, in this circuit to reduce the RC time constant associated with the outputs of the op-amp charging the input capacitance of the op-amp. If, for example, the input capacitance of the op-amp is 25 fF (from parasitics and the MOSFETs used on the op-amp's input), then the RC time associated with charging this capacitance through a 20k resistor is 0.5 ns. As seen in Fig. 26.24, the settling time is approximately 2.5 ns so this RC time can have a significant effect on the settling time and stability of the circuit (important).

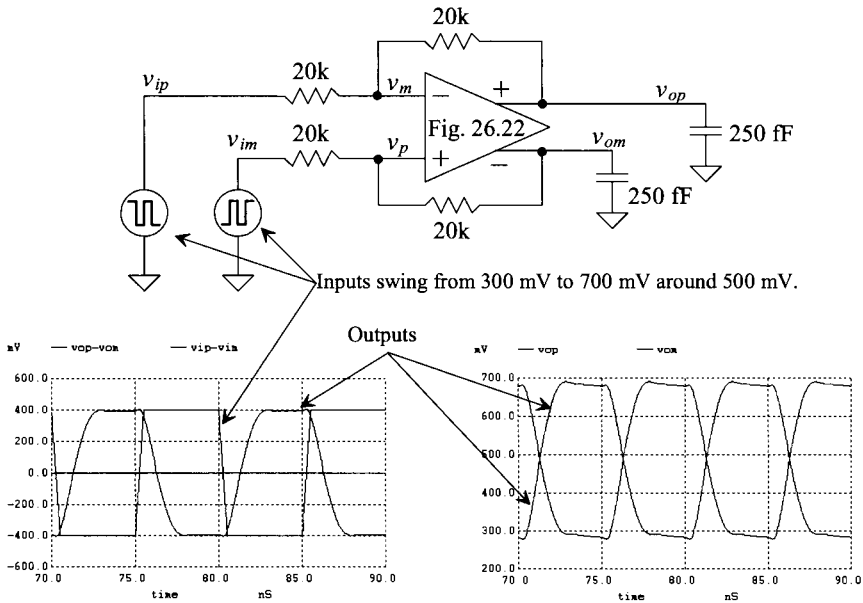


Figure 26.24 Step response of the op-amp in Fig. 26.22 driving 250 fF load capacitors and 20k feedback resistors.

Notice how, in Fig. 26.24, we used input signals that don't swing rail-to-rail. Since we don't have a CMFB circuit in the op-amp, the exact common-mode output voltage is an unknown. It may be 400 mV or it may be 600 mV. We get some help in setting the circuit's output common-mode level by using input signals with common-mode voltages of 500 mV and DC feedback. Looking at the simulation results in Fig. 26.24, we might get a false sense of not needing a CMFB circuit. To illustrate this is indeed a false sense, consider the sample-and-hold seen in Fig. 26.25 (see also Fig. 25.19). When the ϕ switches are closed, the op-amp's inputs and outputs should be held to $V_{CM} \pm V_{OS}$ (the op-amp is placed in the follower configuration). As seen in the figure, the outputs during this time are driven close to 400 mV. On closer inspection, we see the output common-mode level is wandering downwards until eventually the op-amp shuts off.

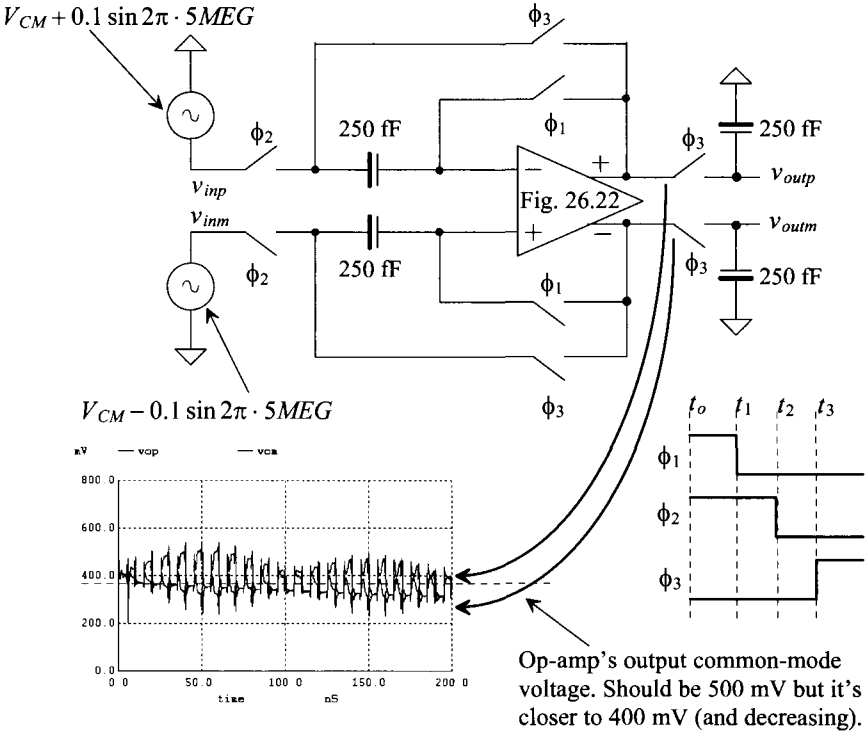


Figure 26.25 A sample-and-hold circuit. Notice how the output common-mode voltage is wandering.

Adding CMFB

Figure 26.26 shows how we can modify the basic op-amp to allow for a CMFB signal input. If the outputs of the op-amp (their average or common-mode voltage) are too high,

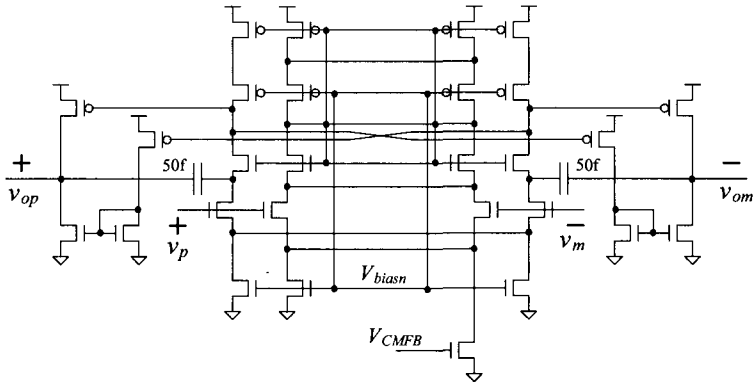


Figure 26.26 Modifying the op-amp for a CMFB input signal.

V_{CMFB} goes up. This causes the output voltage of the diff-amp to go up (increasing V_{CMFB} causes the bias current in the center MOSFETs of the diff-pair to increase). An increase in the diff-amp's output voltage lowers the quiescent current flowing in the output buffers, causing the output voltage to move downwards. This is a good time to remember one of the fundamentals from Ch. 20, namely, if two MOSFET gate-source voltages are equal and they have the same drain current, then their drain-source voltages must be equal. For the output buffer in Fig. 26.26, this means that as we reduce the drain currents flowing in the output buffer (by driving V_{CMFB} high), the gate-source voltages of the NMOS devices decrease. Because of the fundamental concept just mentioned, this causes the output voltages of the op-amp to decrease. Figure 26.27 shows the op-amp's output voltage change with V_{CMFB} . For stability concerns, it's of interest to determine the gain from the CMFB input to the outputs. From the simulation, the gain is approximately 25 (about 10 times less than the differential gain). The lower forward CMFB gain allows us to use a diff-amp with a current mirror load for the CMFB amplifier (discussed next).

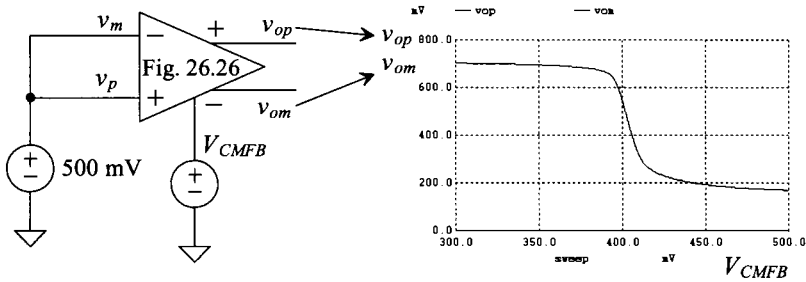


Figure 26.27 The CMFB input to output relationship. The gain is approximately 25 (considerably less than the forward differential gain).

CMFB Amplifier

Now that we know how our CMFB signal, V_{CMFB} , in the op-amp of Fig. 26.26 affects the output voltages and that the gain this signal sees is approximately 25 ($v_{op,n}/V_{CMFB}$), we need to discuss the CMFB amplifier. Consider the two diff-amps seen in Fig. 26.28. In the top diff-amp, gate-drain-connected loads are used. As seen in the simulations, the gain from the averaged input, V_{CMA} , to the diff-amp's output, V_{CMFB} is considerably less than 1 (and so the CMFB is guaranteed to be stable). Unfortunately, the CMFB amplifier's output voltage isn't high enough. (As seen in Fig. 26.27, we need approximately 400 mV.) The diff-amp's tail current can be increased in size (use at least two PMOS for the tail current) to increase the output voltage. However, the low gain, say around 0.3, combined with the CMFB gain through the op-amp, again around 25, means that the CMFB loop's overall gain is only around 7. This isn't large enough to precisely balance the outputs. Using the diff-amp with current mirror load, Fig. 26.28b, gives a gain of approximately 8. This combined with the op-amp's CMFB gain gives an overall CMFB loop gain of 200. This is less than the op-amp's differential gain (so we can use the same compensation capacitors to stabilize the CMFB loop), Fig. 26.23, and large enough to precisely balance the op-amp's outputs.

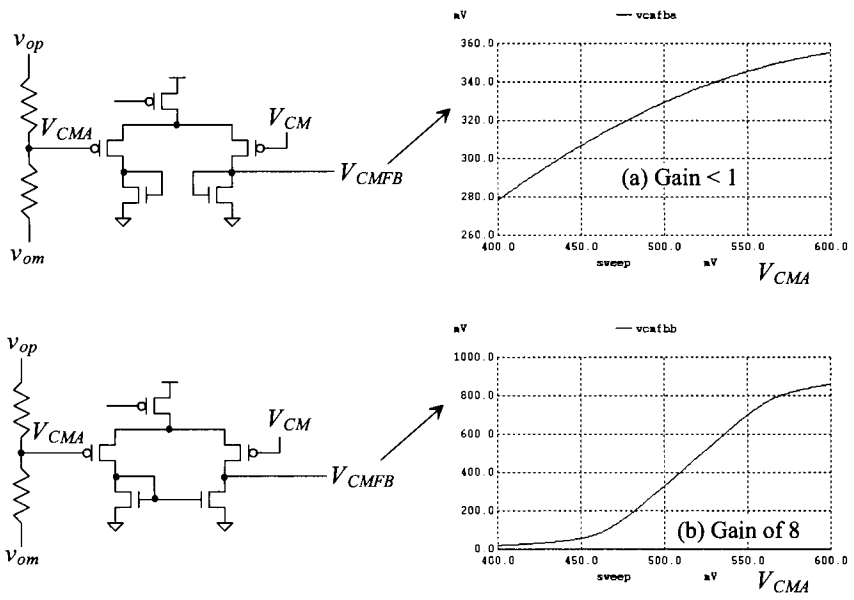


Figure 26.28 Gains of CMFB amplifiers.

The Two-Stage Op-Amp with CMFB

Figure 26.29 shows the complete schematic of the op-amp. Notice that we've doubled the width of the MOSFETs in the output buffer. The added loading from the CMFB averaging resistors will lower the open-loop gain. To compensate for this reduction, the

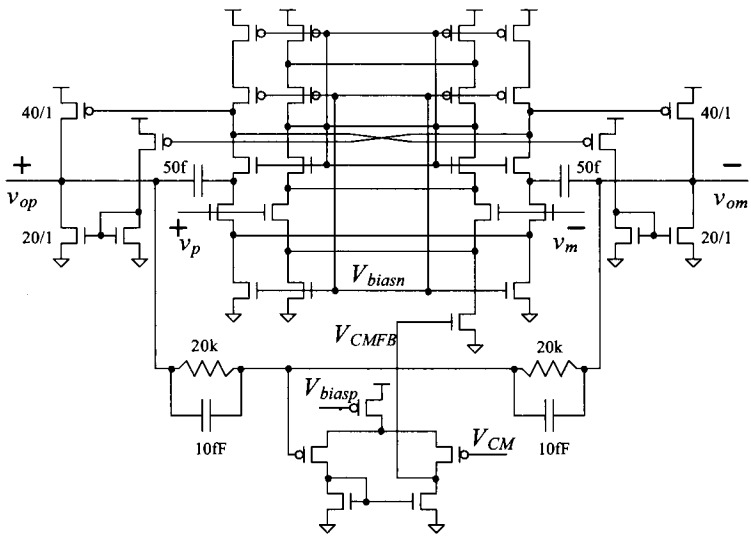


Figure 26.29 Complete schematic of op-amp with CMFB.

drive strength of the MOSFETs on the output of the op-amp was increased. Figure 26.30 shows the DC behavior of the op-amp when placed in the configuration seen in Fig. 26.23. Notice that the outputs cross at the ideal common-mode voltage of 500 mV. The gain is reduced because of the loading by the 20k resistors in the CMFB circuit (and so the gain can be increased by increasing the values of the resistors in the CMFB circuit). However, **we see a problem** with these simulation results. The outputs are only swinging from 200 to 800 mV (not from 0 to V_{DD} as possible with a push-pull output stage). Further, we can estimate the current pulled from V_{DD} for the op-amp in Fig. 26.29 by counting the number of branches in the op-amp (noting that the output stage counts twice because we've doubled the widths of these devices). Including the CMFB amplifier, there are 11 branches. If the bias current through each branch is 20 μA and the current pulled by the bias circuit is 50 μA , then we would expect the op-amp to pull 270 μA (perhaps even a little less because many of the MOSFETs are biased near the triode region). When we look at the simulation results that generated Fig. 26.30, we see that the current is closer to 500 μA (way off indicating, again, that we've got a problem).

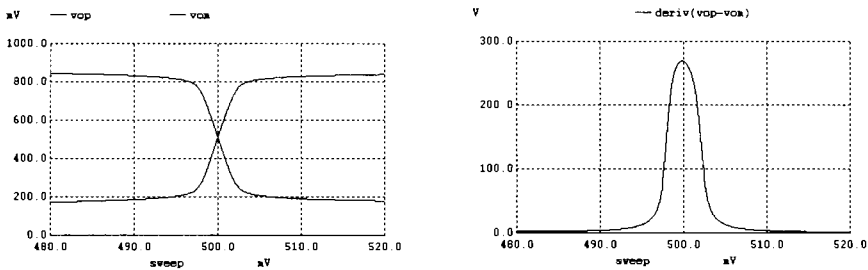


Figure 26.30 Simulating the operation of the op-amp in Fig. 26.29.

Origin of the Problem

The origin of the problem (that is, the op-amp drawing too much current and the output swing not reaching the rails) can be traced to the output buffer. We spent a considerable amount of time discussing how we want to set the outputs of the diff-amp to V_{biasp} . When we look at the simulations, we see the diff-amp's outputs are considerably less than the ideal 600 mV of V_{biasp} . To understand why, let's look at the output buffer in Fig. 26.31. Since the problems appear with the addition of the CMFB circuit, we assume that the two gates of the PMOS devices are moving at the same potential (tied together), that is, that the inputs to the buffer are moving with the diff-amp's output common-mode level. When the output of the diff-amp is V_{biasp} (600 mV), 20 μA flows in all of the MOSFETs. (To keep things simpler, we don't include the doubling in the widths used in the op-amp output devices.) The gate potentials of the NMOS are, roughly, V_{biasn} (400 mV). Because of the symmetry of the circuit, this means that the output is also at 400 mV. To increase the output voltage to 500 mV, we must drop the potential on the gates of the PMOS until the gate-drain-connected NMOS has a V_{GS} of 500 mV (again because of the symmetry). This increases the current (significantly) flowing in the output buffer, lowers its gain, and reduces the linear output swing. The overdrive voltages essentially change from the desired 100 mV when 20 μA of current flows to 200 mV when the output is driven to 500 mV.

Simulation Results

Figures 26.34 to 26.36 show simulation results based on the circuit topologies in Figs. 26.23 to 26.25, respectively, using the op-amp in Fig. 26.33. In Fig. 26.34 we see that the outputs now swing close to the power supply rails (unlike what we saw in Fig. 26.30). Also, the gain is higher. Figure 26.35 shows the step response of the amplifier in the topology seen in Fig. 26.24. Finally, Fig. 26.36 shows the outputs of the sample-and-hold in Fig. 26.25 using the op-amp in Fig. 26.33.

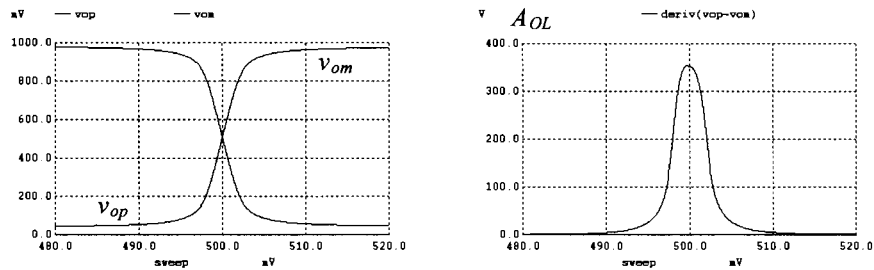


Figure 26.34 Resimulating the op-amp in Fig. 26.33 in the configuration seen in Fig. 26.23.

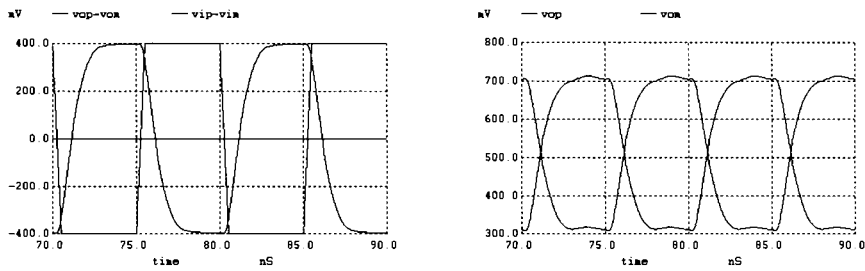


Figure 26.35 Regenerating the simulation results using the topology in Fig. 26.24 with the op-amp in Fig. 26.33.

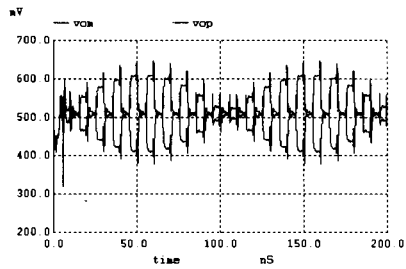


Figure 26.36 Using the op-amp in Fig. 26.33 in the sample-and-hold circuit of Fig. 26.25. Figure shows the outputs of the op-amp.

Using MOSFETs Operating in the Triode Region

For the output buffer in Fig. 26.32, we added a transistor in the drain portion of the circuit. We might wonder if we can accomplish better control of the current flowing in the output buffer by adding circuitry to the source side of the buffer. Towards answering this question, consider the portion of the output buffer seen in Fig. 26.37. If both outputs are at V_{CM} , the gate-source voltages of M2 and M3 are the same and so are the gate-source voltages of M1 and M4. If one output goes high and the other output goes low (and the outputs are centered around V_{CM}), then the net drain current of each M3 is constant. If one output goes below the threshold voltage of an NMOS device, then the balancing stops working. The practical problem of using triode-operating MOSFETs (in a CMFB circuit or any type of amplifying configuration) is that the gain through a MOSFET operating in the triode region is low (so it's difficult to provide control).

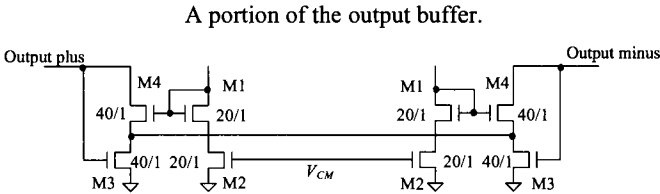


Figure 26.37 Using triode-operating MOSFETs to balance the outputs (bad).

Start-up Problems

Examine the circuit in Fig. 26.38 using the op-amp in Fig. 26.33. Because the op-amp's inputs are at 0 V, the diff-amps on the input of the op-amp are off. This causes the gates of the PMOS devices in the output buffer to be pulled to V_{DD} . The circuit remains in this state and doesn't move the outputs or the inputs up to V_{CM} . To avoid this start-up problem, we need to ensure that there is some DC path to V_{DD} or V_{CM} to "start-up" the op-amp.

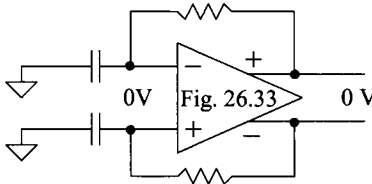


Figure 26.38 The op-amp in Fig. 26.33 won't turn on in this topology.

Lowering Input Capacitance

Consider connecting the gates of the NMOS diff-amp used for biasing to V_{CM} (see the bold line in Fig. 26.39). This reduces the input capacitance of the op-amp and shouldn't affect the normal operation as long as the common-mode voltage of the op-amp (the gates

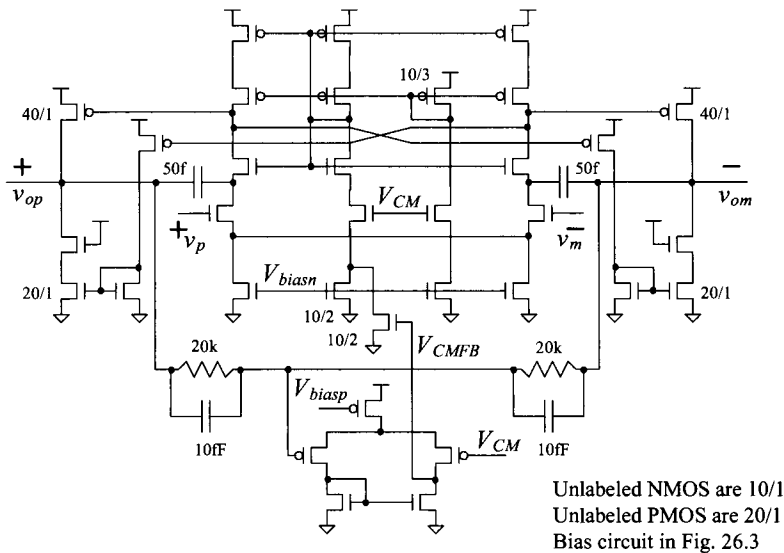


Figure 26.40 Making the op-amp more practical.

A more practical problem is the change in the MOSFET's drain currents with drain-source voltages. For example, looking at Figs. 26.18a and b, we see that it's possible for the NMOS's drain current to be twice the PMOS's drain current for the same gate-source voltages. This can cause op-amp failure in the CMFB circuit. If, for example, the current flowing in the 10/2 MOSFET connected to V_{biasn} is larger than the current sourced by the PMOS, the voltage V_{CMFB} goes to zero and the CMFB loop doesn't work properly. A good "rule-of-thumb" is for the fixed current flowing in the CMFB-controlled bias circuit to be 25–50% of the total expected current.

Increasing the Op-Amp's Open-Loop Gain

The op-amp that we've developed in this section has an open-loop gain in the hundreds. As we'll see in Ch. 29 (Eq. [29.59]), the open-loop gain of the op-amps used in a data converter has a direct effect on the maximum attainable resolution. Towards increasing the gain of the op-amp, let's use the gain-enhancement (GE) techniques presented in Sec. 24.4.

Figure 26.41 shows how we should *not* implement GE. An amplifier with fully-differential outputs, like the amplifier in Fig. 26.7, regulates the drains of the top PMOS devices. We know that this is bad because we would need a CMFB circuit to balance the outputs of the added amplifier.

Figure 26.42 shows how we can add GE to the op-amp developed in this section. Notice that GE is implemented with amplifiers having single-ended outputs. When designing the added amplifiers, as discussed in Sec. 24.4, the bandwidth isn't important for high-speed operation. The lengths of the MOSFET in the added amplifiers can be increased to reduce power and boost gain. An important concern is the added amplifier's input common-mode range.